

Yoshitaka Inoue

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Machine Learning for Biological Perturbations and Therapeutic Response

University of Minnesota | Computer Science and Engineering
National Institutes of Health | NLM / NCI

inoue0426.github.io
Google Scholar | ORCID | GitHub

Research interests: Machine learning for biological perturbations and therapeutic response; single-cell and functional genomics; intervention-aware representation learning; graph-based biomedical reasoning; LLM-based scientific reasoning.

Education

2021–present	University of Minnesota, Twin Cities <i>PhD Candidate, Computer Science and Engineering</i> Research on treatment-conditioned perturbation modeling, drug-response prediction, single-cell and functional genomics, graph representation learning, intervention-aware representations, and LLM-based biomedical reasoning. Advisor: Prof. Rui Kuang; Co-advisor: Dr. Augustin Luna (NIH). Expected graduation: September 2027.
2017–2019	Nara Institute of Science and Technology <i>M.Eng., Information Science</i> Computational prediction and annotation of small molecules from tandem mass spectrometry (MS/MS) data in collaboration with the UC Davis Genome Center and DataLab. Advisor: Prof. Shigehiko Kanaya.
Summer 2018	University of California, Davis <i>Visiting Scholar, Genome Center and DataLab</i> Collaborative research on MS/MS-based compound prediction and metabolomics data analysis. Advisors: Prof. Oliver Fiehn and Tobias Kind.
2012–2017	Kumamoto University <i>B.Sc., Biological Science</i> Experimental analysis of seagrass metabolism using molecular biology methods, including Northern blotting. Advisor: Prof. Susumu Takio.

Research and Professional Experience

2024–present	National Library of Medicine / National Cancer Institute, NIH <i>Pre-doctoral Fellow</i> Developing machine learning methods for perturbation modeling and therapeutic response prediction, including treatment-conditioned latent-state transition models, graph-based drug-response models, and biomedical reasoning systems across single-cell, pharmacogenomic, high-throughput screening, cancer cell-line, and patient datasets.
2022–2024	Google Summer of Code <i>Contributor (2022); Mentor (2023–2024)</i> Developed a drug-response prediction pipeline and mentored projects on PyTorch Geometric datasets and multi-tool LLM aggregation for biomedical analysis.
2019–2021	FreakOut, Inc. <i>Machine Learning Engineer</i> Applied machine learning to click-through-rate prediction in Tokyo, Japan.

Selected Publications

- Inoue, Y., Jeong, M., Hero, A., Kuang, R., & Luna, A.**
PerturbRx: Learning Treatment-Conditioned Latent Transitions for Patient Drug Response Prediction.
arXiv, 2026; submitted to ICLR 2027
- Inoue, Y., Song, T., Wang, X., Luna, A., & Fu, T.**
DrugAgent: Reliable Multi-Agent Aggregation under Conflicting Biomedical Evidence.
Scientific Reports, 2026, under revision
- Inoue, Y., Lee, H., Fu, T., Kuang, R., & Luna, A.**
drGT: Attention-Guided Gene Assessment of Drug Response Utilizing a Drug-Cell-Gene Heterogeneous Network.
BMC Bioinformatics, 2026
- Inoue, Y., Hao, N., Lu, Y., Fu, T., & Luna, A.**
Drug Discovery in the Era of Artificial Intelligence: From Target Identification to Clinical Trials.
Health Data Science, 2026, accepted

Selected Publications (continued)

5. Hoshi-Kadonosawa, Y., Reinhold, W. C., Taniyama, D., **Inoue, Y.**, et al.
TRPM4 Expression as a Predictive Biomarker and a Mechanistic Driver of Acetalax Activity in Prostate Cancer: Preclinical Efficacy Studies.
Molecular Cancer Therapeutics, 2026

6. Elloumi, F., Reinhold, W. C., Varma, S., Wang, Y., Kinali, M., Arakawa, Y., **Inoue, Y.**, et al.
CellMiner cross-database (CellMinerCDB) version 2.2 for explorations of patient-derived cancer cell line pharmacogenomics.
Nucleic Acids Research, 2026

7. Taniyama, D., **Inoue, Y.**, Elloumi, F., Kim, Y. S., et al.
Spontaneous dynamic gene expression of Schlafen 11 (SLFN11) demonstrated by single-cell RNA sequencing in clonal cancer cell lines.
Oncogene, 2026, submitted

Conference and Workshop Papers

1. **Inoue, Y.**, Song, T., Wang, X., Luna, A., & Fu, T.
DrugAgent: Reliable Multi-Agent Aggregation under Conflicting Biomedical Evidence.
ICML Workshop on Foundation Models for the Life Sciences (FM4LS), 2026

2. **Inoue, Y.**, Fu, T., & Luna, A.
GraphPINE: Graph Importance Propagation for Interpretable Drug Response Prediction.
ICLR Workshop on Machine Learning for Genomics Explorations (MLGenX), 2025

3. **Inoue, Y.**, Song, T., Wang, X., Luna, A., & Fu, T.
DrugAgent: Explainable Drug Repurposing Agent with Large Language Model-based Reasoning.
ICLR Workshop on Machine Learning for Genomics Explorations (MLGenX), 2025

Preprints and Research Software

1. **Inoue, Y.**, Kulman, E., & Kuang, R.
BiGCN: Leveraging Cell and Gene Similarities for Single-cell Transcriptome Imputation with Bi-Graph Convolutional Networks.
bioRxiv, 2024

2. **OncoGraph**: open oncology knowledge graph and retrieval benchmark integrating drugs, genes, diseases, clinical trials, Gene Ontology terms, and provenance-linked biomedical evidence. github.com/inoue0426/OncoGraph

Selected Presentations

International Conference on Intelligent Systems for Molecular Biology (ISMB)		2026
Two accepted oral presentations: drGT in NetBio and DrugAgent in Text Mining; both also presented at the Student Council Symposium.		Washington, DC
26th Annual CCR-FYI Colloquium		2026
drGT oral presentation; selected as one of four Outstanding Postgraduate Fellow Finalists		Rockville, MD
NIH Artificial Intelligence Symposium		2025
drGT: attention-guided gene assessment for drug-response modeling; selected among 5 of 78 listed abstracts		Rockville, MD

Honors and Fellowships

- **NIH Pre-Doctoral Research Fellowship**, 2024–present.
- **Outstanding Postgraduate Fellow Finalist**, 26th Annual CCR-FYI Colloquium, 2026; selected as 1 of 4 finalists.
- **Research Visit Scholarships**, NAIST and JASSO support for UC Davis research visits, 2018–2019.

Academic Service and Memberships

Journal reviewing	BMJ Digital Health & AI (Expert Peer Reviewer, 2026–2027); IEEE/ACM Transactions on Computational Biology and Bioinformatics (2025–2026); Nature Computational Science (co-reviewer, 2026).
Conference reviewing	ACM BCB (2023); ICML FM4LS (2026); ICML GenBio (2026); IEEE BIBM (2026); ACM SIGSPATIAL (2026).
Memberships	International Society for Computational Biology (ISCB); American Association for Cancer Research (AACR); American Society of Clinical Oncology (ASCO).

Technical Skills

ML / AI	Perturbation and transition modeling; graph neural networks; representation learning; LLM-based agents; interpretable machine learning.
Computational biology	Pharmacogenomics; functional genomics; biological networks; single-cell RNA-seq; therapeutic response modeling; multi-omics analysis.
Programming	Python; PyTorch; PyTorch Geometric; scikit-learn; pandas; NumPy.
Infrastructure	Git; Linux; Jupyter; Slurm; GPU computing; HPC environments; LangChain.